# Shrijani Vemulapally

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## Education

**Cornell University** August 2026 – May 2027
*Master of Engineering, Computer Science*
- **Relevant Coursework:** Robot Perception | Systems for Large-Scale Machine Learning | Systems Programming | Frontiers of Computer Vision
- **Teaching Assistant, CS 2110 (Java):** Lead discussions and office hours; coordinate assignment grading and resolve rubric inconsistencies for a 300+ student course.

**University of California, Irvine** December 2024
*B.S. Computer Science & Engineering* — ***GPA: 3.8***

## Relevant Experience

**Software Development Engineer, Amazon** February 2025 – August 2026
*Health Services, Growth and Marketing Engagement (GAME)* *Seattle, WA*
- Improved widget show rate approximately 4x at constant CTR by implementing weekly ID rotation in Java to trigger renewed exploration by an external ranking system.
- Architected an asynchronous EventBridge, Lambda, and Bedrock recommendation pipeline, partnering with the PM on requirements and design; kept LLM inference off infrastructure's 6,200-peak-RPS request path.
- Engineered internal AI agents using MCP integrations and a code/docs knowledge base to automate code reviews, design drafts, and implementation, freeing the team to focus on architecture.
- Implemented LLM-as-judge checks for PHI exposure, unsupported health assumptions, question length, and clinical complexity to evaluate generated recommendations.
- Drove A/B experiments with BI teams, defining metrics and dashboards to evaluate a feature that increased customer acquisition by 20%.
- Shipped customer-facing React/TypeScript widgets on Amazon's shopping portal, including video modules, recommendation carousels, and feeder pills.

**Software Development Engineer Intern, Amazon** June 2024 – September 2024
*Health Services* *Seattle, WA*
- Developed a RAG prototype for an Amazon Health team of 10–15 Product Managers to explore customer-feedback insights through natural-language queries.
- Automated S3-triggered ingestion in SageMaker: chunked data, generated Titan embeddings, and stored them in DocumentDB for KNN retrieval and Claude 3.5 Sonnet answers.

## Projects

**ContextIQ** — Alexa+ Memory Agent | Amazon Developers Hackathon | *Best Design, 70+ teams* April 2026 – May 2026
- Prototyped an Alexa+ memory layer that retrieves past Bee Pioneer conversations to resolve underspecified requests, reducing follow-up questions by up to five.
- Designed Bedrock/Neptune Graph RAG retrieval combining semantic search and graph expansion; reranked 15 retrieved chunks to the top five with Cohere Rerank 3.5.
- Built a voice agent on Amazon Nova Sonic and the Strands SDK that executes context-aware actions across 8 tools spanning shopping, reservations, calendar, and task management.

**Pixory** — iOS Photo Storytelling App June 2026 – Present
- Built an iOS photo-storytelling app using React Native/Expo, Node.js/Fastify, and Python computer vision; curates 300 trip photos into stories for six private-beta users.
- Engineered a 300-to-120-to-30 photo-selection pipeline using time/location buckets and nine visual scores before GPT-4o selects, sequences, and captions the story.
- Implemented edit-driven story regeneration and explanations; learned scoring weights from user swaps to personalize photo selection after five uses.

**Labmate** — Local Research Agent | DELL x NVIDIA Hackathon September 2026
- Collaborated in a four-person team on a local research agent; developed Hermes skills and autonomy with sandboxed Qwen inference for unpublished research.
- Developed deterministic Python/pandas claim checks and a file watcher that rechecks affected claims when source data changes, preserving evidence locally.

## Technical Skills

**Languages:** Python | Java | TypeScript | C++ | Rust
**AI & Data:** LLM Applications | RAG | Graph RAG | LangChain | LLM Evaluation | Vector Search | Agentic Workflows
**Backend & Cloud:** AWS | Bedrock | CDK | REST APIs | Event-Driven Architecture | Distributed Systems | PostgreSQL
**Frontend & Testing:** React | React Native | Expo | Playwright | Mockito | Unit Testing | Integration Testing